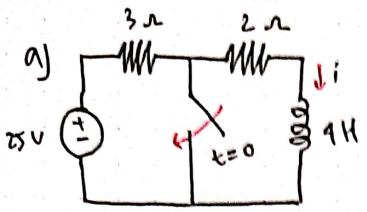
NAMA : RAY ERLANGGA R
NIM : 20133105S
Kelas : 1B - TTE

(1) A. Tentukan nilai arus dan tegangan di sisi induktor, untuk $t>0$

a)

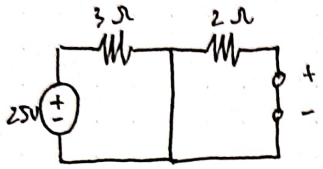


* untuk $t>0$ saklar tertutup (short circuit)

$\tau = \frac{L}{R}$ $\quad i(t) = 5e^{-\frac{t}{\tau}}$

$\tau = \frac{9}{2}$ $\quad i(t) = 5e^{-\frac{t}{2}}$ A

$\tau = 2$



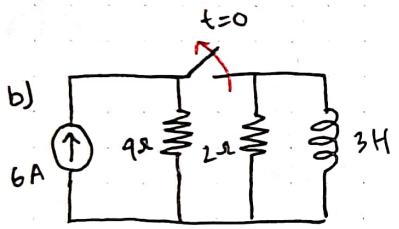
$V(t) = L\frac{di}{dt}$

$= 9\left(-\frac{5e^{-\frac{t}{2}}}{2}\right)$

$= 9\left(-2{,}5\,e^{-\frac{t}{2}}\right)$

$= -10\,e^{-\frac{t}{2}}$ V

b)



* untuk $t>0$ saklar terbuka (open circuit)

$\tau = \frac{L}{R}$ $\quad i(t) = 6e^{-\frac{t}{\tau}}$

$\tau = \frac{3}{2}$ $\quad i(t) = 6e^{-\frac{t}{3/2}}$

$i(t) = 6e^{-\frac{2t}{3}}$ A

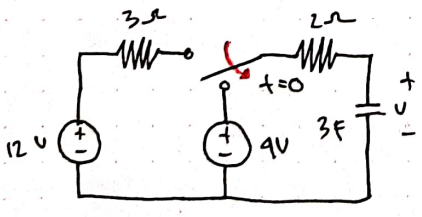
$V(t) = L\frac{di}{dt}$

$= 3\left(-\frac{6e^{-\frac{2t}{3}}}{3/2}\right)$

$= 3\left(-\frac{6e^{-\frac{2t}{3}}}{1{,}5}\right)$

$= 3\left(-4e^{-\frac{2t}{3}}\right) = -12\,e^{-\frac{2t}{3}}$ V

B. Tentukan nilai arus (i) dan tegangan (v) di kapasitor



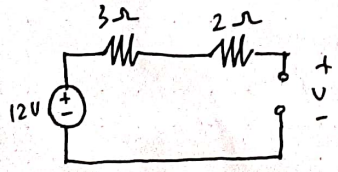
Dik : $V_{s1} = 12$V $\quad C = 3$F

$V_{s2} = 9$V

$R_1 = 3\Omega$

$R_2 = 2\Omega$

* untuk $t<0$ saklar terhubung dengan $R = 3\Omega$.



$V(0^+) = V(0^-)$

$V(0) = V(0^+) = V(0^-)$

$V(0) = 12$ V

Ketika $t<0$ $V(t) = 12$ V

$V = I.R$

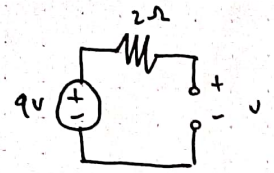
$12 = I.(3+2)$

$12 = I.(5)$

$I = \frac{12}{5} = 2{,}4$ A

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* untuk $t>0$ saklar terhubung dengan $V_{S2} = 9\,V$



$\tau = R \cdot C$

$\tau = 2 \cdot 3$

$\tau = 6$

$$v(t) = v_{(\infty)} + (v_{(0)} - v_{(\infty)})\, e^{-\frac{t}{\tau}}$$

$$= 9 + (12 - 9)\, e^{-\frac{t}{6}}$$

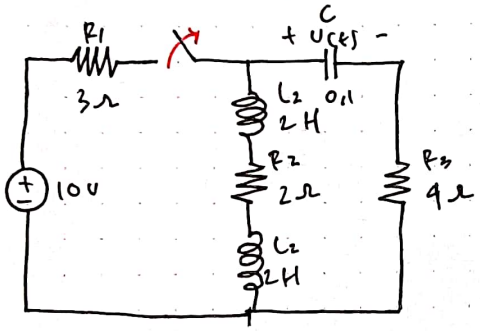
$$= 9 + 8\, e^{-\frac{t}{6}}\, V$$

$$i(t) = C \frac{dv}{dt}$$

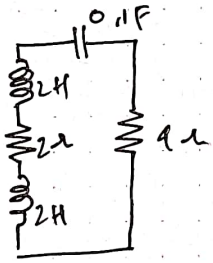
$$= \tfrac{1}{3}\left(-\frac{9e^{-t/6}}{2}\right)$$

$$= -9e^{-\frac{t}{6}}\, A$$

(2) tentukan nilai $v(t)$ dan tegangan di R_2, untuk $t>0$



* untuk $t>0$ (terbuka)



$L_p = 2+2 = 4\,H$

$$\alpha = \frac{2+9}{2.9} = \frac{6}{8} = 0{,}75\ rad/s$$

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{9.0{,}1}} = 1{,}58\ rad/s$$

$\alpha < \omega_0 \rightarrow$ rangkaian underdamped

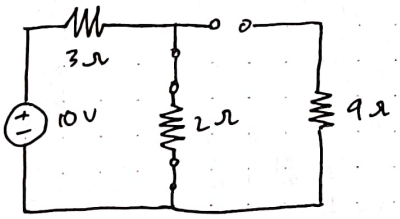
$$v(t) = (A_1 \cos \omega t + A_2 \sin \omega t)\, e^{-\alpha t}$$

$t = 0$

$9V = (A_1 + 0)\, e^{-0}$

$9V = A_1$

* untuk $t<0$ (tertutup)



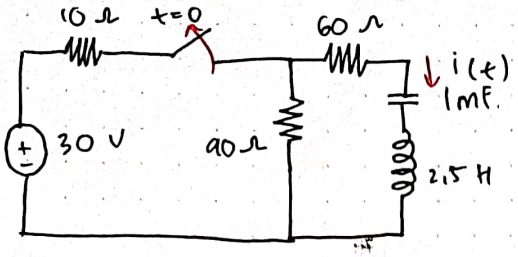
$$V(t) = V_{2\Omega} = 10 \times \frac{2}{3+2}$$

$$= 10 \times \frac{2}{5}$$

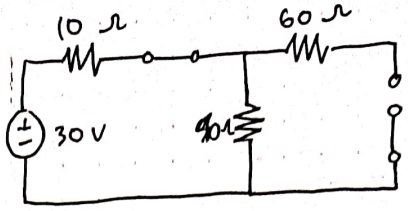
$V(0) = 9\,V$

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(3) tentukan nilai arus $i(t)$ dan tegangan disisi induktor untuk $t>0$



* untuk $t<0$

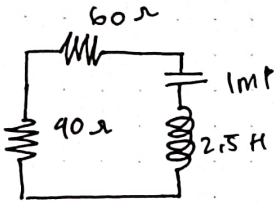


$$V_{(0)} = \frac{90}{10+90} \cdot 30$$

$$V_{(0)} = \frac{90}{50} \cdot 30$$

$$V_{(0)} = 29 \text{ V}$$

* untuk $t>0$



$$\alpha = \frac{R}{2L} = \frac{90+60}{2(2,5)} = 20$$

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{(2,5)(1\times10^{-3})}} = \frac{1}{\sqrt{2,5\times10^{-3}}} = 20$$

$\alpha = \omega_0$, maka rangkaian menjadi critically damped case

* $i(t) = (A_2 + A_1 t)e^{-\alpha t}$

$i(0) = A_2 = 0$

* $\frac{di}{dt} = \frac{d}{dt}(A_1 t e^{-20t})$

$= A_1 e^{-20t} - 20A_1 t e^{-20t}$

* $\frac{di(0)}{dt} = -\frac{1}{L}(R I_0 + V_0)$

$= -\frac{1}{2,5}(0+29)$

$= -9,6$

$A_1 e^{-20(0)} - 20 A_1(0) e^{-20(t)} = -9,6$

$A_1 = -9,6$

$i(t) = -9,6 t e^{-20t}$ A

* $V_L(t) = L\frac{di}{dt}$

$= 2,5(A_1 e^{-20t} - 20 A_1 t e^{-20t})$

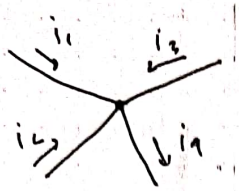
$= 2,5 A_1 e^{-20t} - 50 A t e^{-20t}$

$= 2,5(-9,6)e^{-20t} - 50(-9,6)te^{-20t}$

$V_L(t) = -24e^{-20t} + 480 t e^{-20t}$ V

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④ A. Tentukan nilai i_4



i. $i_1 = 6\cos 3t$ A
$i_2 = 9\cos(3t - 30°)$ A
$i_3 = -9\sqrt{3}\cos(3t + 60°)$ A

ii. $i_1 = 5\cos(3t + 30°)$ A
$i_2 = 5\sin 3t$ A
$i_3 = 5\cos(3t + 150°)$ A

(i) $i_1 = 6\angle 0°$ A
$i_2 = 9\angle -30°$ A
$i_3 = -9\sqrt{3}\angle 60°$ A

KCL :

$-i_1 - i_2 - i_3 + i_4 = 0$

$i_4 = i_1 + i_2 + i_3$
$i_4 = 6\angle 0° + 9\angle -30° + (-9\sqrt{3}\angle 60°)$
$i_4 = 10\angle -53{,}13°$

$i_4 = 10\cos(3t - 53{,}13°)$ A

(ii) $i_1 = 5\angle 30°$ A
$i_2 = 5\cos(90 - 3t)$ A $= 5\cos(3t - 90°) = 5\angle -90°$ A
$i_3 = 5\angle 150°$ A

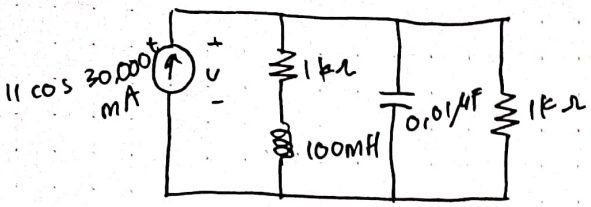
KCL :

$-i_1 - i_2 - i_3 + i_4 = 0$

$i_4 = i_1 + i_2 + i_3$
$i_4 = 5\angle 30° + 5\angle -90° + 5\angle 150°$
$i_4 = 0$ A

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B. tentukan besar v dan arus yg mengalir pada C = 0,01 μF



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